The Cowan Atomic-Structure Codes

What RCN, RCN2, RCG, RCE Actually Compute, from First Principles

A working explainer for the Modern Atomic Linelists (`mal`) project

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Abstract

This document explains, from first principles, what the four Cowan programs do and how they chain together to produce atomic energy levels and oscillator strengths (gf-values) — the data that go into stellar line lists. It maps the physics onto the *actual* programs, input decks, and output files in this repository (`cowan_nist/extracted/`). The goal is a shared, durable understanding of the engine we are modernizing, with particular attention to the least-squares fitting stage (RCE), which is the heart of the project. The canonical reference throughout is R. D. Cowan, *The Theory of Atomic Structure and Spectra* (Univ. California Press, 1981), cited as **TASS**.

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1 The big picture: why these four programs exist

A stellar line list needs, for each transition, the lower and upper energy levels, the wavelength, and the oscillator strength gf. The wavelength is just the level difference; the levels and the gf are the hard part. Computing them *ab initio* for complex atoms (open d- and f-shells, the iron group) is not reliably accurate. Cowan's insight, embodied in these codes, is a *semi-empirical* scheme: compute the atomic structure approximately from theory, then *adjust a small number of physically meaningful radial parameters until the predicted levels match the experimentally measured ones.* The wavefunctions (eigenvectors) that result are far better than the purely theoretical ones, and the gf-values computed from them are correspondingly better.

The pipeline in one breath

RCN solves for the radial wavefunctions and the *radial integrals* $(F^k, G^k, \zeta_{n\ell}, \dots)$ of each electron configuration $\rightarrow$ RCN2 forms the configuration-interaction and electric-multipole (transition) radial integrals between configurations $\rightarrow$ RCG builds the angular part, assembles and *diagonalizes* the energy (Hamiltonian) matrix to get energy levels and eigenvectors, and from the eigenvectors computes gf-values $\rightarrow$ RCE re-runs the diagonalization inside a *least-squares loop*, varying the radial parameters until the computed levels best fit the observed levels.

The division of labor reflects a deep structural fact about atomic Hamiltonians: the energy of a level separates into **radial** factors (integrals over the one-electron radial wavefunctions — these are the F^k, G^k, $\zeta_{n\ell}$, R^k) multiplied by **angular** factors (pure angular-momentum algebra, the same for any atom with a given configuration). RCN/RCN2 compute the radial side; RCG computes the angular side and combines them; RCE treats the radial numbers as adjustable.

2 The physics each stage implements

2.1 The central-field starting point and rcn

We seek the eigenstates of the non-relativistic many-electron Hamiltonian

$$H = \sum_i \left(-\tfrac{1}{2}\nabla_i^2 - \frac{Z}{r_i} \right) + \sum_{i<j} \frac{1}{r_{ij}} + \sum_i \xi(r_i)\,(\mathbf{l}_i \cdot \mathbf{s}_i), \tag{1}$$

in atomic units, where the last term is the spin–orbit interaction. The *central-field approximation* writes each electron's wavefunction as a product of a radial part $P_{n\ell}(r)/r$ and angular/spin parts. RCN solves the resulting one-electron radial equations self-consistently for a *single configuration* (e.g. $3d^7$, or $3d^6 4p$), using one of several approximations to Hartree–Fock; in practice the Hartree-plus-statistical-exchange (HX) method or center-of-gravity HF is used, optionally with relativistic corrections (the "HFR" / "HXR" options — important for the iron group and heavier).

From the converged radial wavefunctions $P_{n\ell}(r)$, RCN computes the **radial integrals** that parametrize the energy:

$$F^k(n\ell, n'\ell') = \iint P_{n\ell}^2(r_1)\,\frac{r_<^k}{r_>^{k+1}}\,P_{n'\ell'}^2(r_2)\,dr_1\,dr_2 \qquad \text{(direct Slater integral)}, \tag{2}$$

$$G^k(n\ell, n'\ell') = \iint P_{n\ell}(r_1)P_{n'\ell'}(r_1)\,\frac{r_<^k}{r_>^{k+1}}\,P_{n\ell}(r_2)P_{n'\ell'}(r_2)\,dr_1\,dr_2 \qquad \text{(exchange Slater integral)}, \tag{3}$$

$$\zeta_{n\ell} = \tfrac{\alpha^2}{2} \int P_{n\ell}^2(r)\,\frac{1}{r}\frac{dV}{dr}\,dr \qquad \text{(spin–orbit parameter)}, \tag{4}$$

plus E_{av}, the configuration's center-of-gravity energy. These few numbers per configuration are the *radial parameters*. Everything semi-empirical happens by treating them as adjustable later (§2.4).

2.2 Configuration interaction and transition integrals: rcn2

Real atomic states are not pure single configurations; nearby configurations of the same parity and *J mix* (configuration interaction, CI). RCN2 takes the radial wavefunctions from RCN for *several* configurations and computes:

- the **CI Slater integrals** R^k (the r_{ij}^{-1} matrix elements *between* different configurations), which set the strength of the mixing;
- the **electric-multipole radial integrals** $\langle n\ell \,\|\, r \,\|\, n'\ell' \rangle$ (and higher multipoles), the radial part of the transition dipole moment — the seed of every gf-value.

2.3 Angular algebra, diagonalization, and where gf comes from: rcg

RCG supplies the *angular* half. Using angular-momentum (Racah/Wigner) algebra and tabulated coefficients of fractional parentage (the binary "CFP decks" `FOR072/073/074` that a freshly built RCG generates once), it builds the Hamiltonian matrix in a chosen coupling basis. Schematically, every matrix element of (1) is

$$\langle \gamma' \,|\, H \,|\, \gamma \rangle = \sum_{\text{integrals}} \underbrace{c^{(\cdot)}_{\gamma'\gamma}}_{\text{angular coeff. (RCG)}} \times \underbrace{\{E_{\text{av}}, F^k, G^k, \zeta_{n\ell}, R^k\}}_{\text{radial integrals (RCN/RCN2)}}. \tag{5}$$

RCG then **diagonalizes** this matrix for each J. The eigenvalues are the predicted **energy levels**; the eigenvectors give the **composition** of each level (its percentage breakdown over basis states — e.g. "72% $^5D + 18\%\ ^3F + \dots$").

The oscillator strength follows from the eigenvectors. The line strength between levels βJ and $\beta' J'$ is

$$S = \left| \langle \beta J \,\|\, \mathbf{P}^{(1)} \,\|\, \beta' J' \rangle \right|^2, \qquad gf = \frac{8\pi^2 m_e c}{3h} \frac{\sigma}{g} S \;\propto\; \sigma S, \tag{6}$$

where σ is the transition wavenumber and the reduced matrix element of the dipole operator $\mathbf{P}^{(1)}$ is a sum over the RCN2 transition radial integrals weighted by *both* levels' eigenvector components. This is the crucial coupling that drives the whole project: **many lines' gf-values depend on the same few radial parameters through the shared eigenvectors.** A small change in the fit redistributes oscillator strength among coupled lines.

2.4 The semi-empirical least-squares fit: rce

This is the stage we are modernizing, so we treat it carefully. RCG's predicted levels are only as good as the theoretical radial integrals — and those are systematically off (Hartree–Fock overestimates the Slater integrals by ∼10–20%). RCE fixes this empirically. Its job (TASS §16-3 to 16-5):

1. Treat the radial parameters $\mathbf{x} = \{E_{\text{av}}, F^k, G^k, \zeta_{n\ell}, R^k, \dots\}$ as *free*. The angular coefficients in (5) are fixed (precomputed by RCG and passed in the binary file `TAPE2E`).
2. For trial $\mathbf{x}$, build and diagonalize the energy matrix for each J to get computed levels $E_i^{\text{calc}}(\mathbf{x})$.
3. Minimize the weighted least-squares residual against the *experimentally observed* levels E_i^{obs} (from NIST/lab),

$$\chi^2(\mathbf{x}) = \sum_i w_i \left[E_i^{\text{calc}}(\mathbf{x}) - E_i^{\text{obs}} \right]^2, \tag{7}$$

   iterating (Newton/Levenberg-style) until the parameters stop changing (by < 0.03 between cycles) or a cycle cap is hit. Parameters can be held fixed, or tied in fixed ratios (a group forced to scale together).
4. Feed the *fitted* $\mathbf{x}$ back through the diagonalization to get the *improved* eigenvectors — and hence the improved gf-values (via RCG).

Why this is "black magic" — and our opportunity

The fit in (7) is ill-conditioned: parameters are correlated, the problem is non-linear, and which configurations to include and which parameters to free/fix/tie is a matter of hand-tuned judgment with "no strict methodology." This is the only non-automated stage, takes hours to weeks per ion, and has scarcely changed in 50 years. It is exactly where modern techniques — regularization, Bayesian inference with priors, MCMC for *per-line gf uncertainties*, global optimization, and (our novel idea) folding the observed *stellar spectrum* into the objective alongside the energy levels — can make a genuine advance. See the project plan.

3 Mapping to the actual code and files

Program	Source (`for/`)	Computes
RCN	`RCN36K.F`	Single-config. radial wavefns $P_{n\ell}(r)$; radial integrals $E_{\text{av}}, F^k, G^k, \zeta_{n\ell}$
RCN2	`RCN2K.F`	CI integrals R^k; multipole transition integrals $\langle r \rangle$
RCG	`rcg11k.f`	Angular coeffs (CFP); builds & diagonalizes H $\to$ levels, eigenvectors, gf
RCE	`RCE20K.F`	Least-squares fit of radial parameters to observed levels; improved eigenvectors $\to gf$

Table 1: The four programs and what they compute. Built by `build/build.sh`.

Data flow between programs. Each program writes the input for the next, mostly through fixed-name files (a legacy of the punched-card/tape era):

- RCG consumes the binary CFP decks `FOR072/073/074` (generated once by `build/make_cfp.sh`) and writes, among others, `TAPE2E` (binary coefficient matrices for RCE) and `OUTGINE` (a starter for RCE's text input).
- RCE reads `TAPE2E` plus a text deck `INE20` (control info + the *observed* levels — the only thing not produced upstream) and writes `LEVELS1/2/3` (fitted levels and eigenvectors) and `PARVALS` (fitted parameter values, for restarting).

An actual input deck. The example `work/IN36` is an RCN input for Sn VII+ (`IN36.ai` is B III+ with a long Rydberg series). The first line is a fixed-column "control card" (mesh, method flags, convergence tolerances, relativistic/correlation switches); each subsequent line names one configuration. For Sn VII+:

```
200-90 0 2  01.  4.0    5.E-08    1.E-11-2 00090 0 1.0  0.65  0.0 1.00   -6
   50    8Sn7+  4d7              4p6  4D7
   50    8Sn7+  4d65p            4p6  4D6  5P
   -1
```

Here the two configurations $4d^7$ and $4d^6 5p$ (opposite parity — the source and upper levels of E1 transitions) are listed, terminated by `-1`. The columns of this card are documented in `cowan_nist/extracted/RCN_DOC.` we will annotate them in detail when we drive a real run.

4 What "running it" looks like end to end

For a single ion and one parity pair, the canonical chain is

$$\text{RCN} \rightarrow \text{RCN}2 \rightarrow \text{RCG} \rightarrow \text{RCE} \rightarrow \text{RCG}$$

(the final RCG re-run applies the fitted parameters to emit final levels and gf). The seven coupling schemes RCG/RCE support (LS, JJ, intermediate, ...) only change the basis, not the physics; final eigenvectors are reported in LS or JJ for interpretation.

Next steps in the project

1. Drive the shipped example (`IN36`) through the full chain and read each output file together, annotating the real column formats.
2. Reproduce a known Kurucz Fe II gf via the standard chain (validate we replicate the legacy result).
3. Replace the RCE fit (7) with a modern regularized / Bayesian fit, attach per-line gf uncertainties, and (Tier 2) fold in the solar spectrum.

References

- R. D. Cowan, *The Theory of Atomic Structure and Spectra* (Univ. California Press, 1981). [TASS]
- Program documentation shipped with the code: `RCN_DOC.txt`, `RCG_DOC.txt`, `RCE_DOC.txt` (in `cowan_nist/extracted/`).
- A. Kramida, "Cowan Code: 50 Years of Growing Impact on Atomic Physics," *Atoms* **7**, 64 (2019).